METHODOLOGY DOCUMENT

Toyota Manufacturing Process Improvement Using Lean Six Sigma and Machine Learning

1. Define

Prepare a charter with business case, problem, goal, scope, stakeholders, timeline and CTQs. Create SIPOC and agree the operational definition of a defect.

2. Measure

Document data collection, units, sampling, measurement-system adequacy, calibration, missingness, duplicates and baseline defect rate. Cp/Cpk require valid specification limits and a stable process.

3. Analyze

Use Pareto, stratification by line/shift/machine/material, distributions, trends and correlations. Use t-tests, ANOVA or chi-square only when assumptions and sampling design are appropriate. Predictive importance is not causal proof.

4. Improve

Translate verified root causes into countermeasures. Prioritize impact, feasibility, cost and risk. Pilot approved changes and evaluate before/after outcomes using a suitable design. Do not claim savings before verification.

5. Control

Define CTQ, metric, owner, sampling frequency, chart, standard work, audit cadence and reaction plan. Select chart type based on data and subgroup design. Control limits are not specification limits.

6. Machine-learning workflow

Random Forest predicts binary defect risk. Fit imputation, scaling and one-hot encoding inside the training pipeline. Use stratified holdout for the demo; use temporal/group validation when predicting future periods or unseen machines. Report prevalence, precision, recall, F1, ROC-AUC and confusion matrix.

7. Reproducibility and governance

Record data provenance, units, exclusions, random seed, package versions, split strategy, feature list and evaluation date. Protect confidential production and supplier information. Validate actions with quality/process engineers. Do not use model output as an automatic machine-control instruction.

8. Interpretation warning

The bundled dataset is synthetic and not Toyota production data. Company-specific conclusions require authorized data and independent validation.